RAJALAKSHMI ENGINEERING COLLEGE
Approved by AICTE | Affiliated to Anna University | Accredited by NAAC

Register No.: 250701822

Continuous Assessment-II		
Computer Science and Engineering		
Degree: B. E / B. Tech	Semester: III	Year: II
Course code & Title:	CS23331 & Design and Analysis of Algorithms	
Date: 29.10.26		QP CODE: 071104
[Common to: IT, AIML, AI&DS and CSD]		

[Regulations: 2023]

Time: 2 hr 15 min

Max. Marks: 75

Answer All Questions

Part A: 10 x 2 = 20 Marks

1	State the recurrence relation for the number of moves required to solve the Towers of Hanoi problem with *n* disks.	[CO3]	[K1]
2	Compare the behavior of Quick Sort and Merge Sort when the input elements are already sorted.	[CO3]	[K2]
3	Define the optimal substructure property to the applicability of dynamic programming.	[CO4]	[K1]
4	Define the Longest Common Subsequence problem and state one practical application.	[CO4]	[K1]
5	Compare Longest Common Subsequence and Longest Common Substring with respect to the arrangement of matched elements.	[CO4]	[K2]
6	For a staircase that permits steps of size 1 or 2, find the number of distinct ways to reach the fourth step.	[CO4]	[K3]
7	Compare backtracking and branch-and-bound with respect to the criterion used for pruning the search space.	[CO5]	[K2]
8	A graph contains a triangle formed by vertices A, B, and C. Find the minimum number of colors required for a valid graph coloring.	[CO5]	[K3]
9	Define an NP-Complete problem and state one example.	[CO5]	[K1]
10	Compare NP-Complete and NP-Hard problems with respect to membership in the class NP.	[CO5]	[K2]

Part B: 5x11=55 Marks

11 a. Consider the following array: [CO3] [K3]

Index: 0 1 2 3 4 5 6

| 42 | 18 | 35 | 27 | 50 | 12 | 31 |

Pivot = 31

Apply Quick Sort to sort the given elements using the first element as the pivot. Analyze the best-case and worst-case time complexity of Quick Sort

[OR]

b. Consider the following sequence: [CO3] [K3]

[38, 16, 27, 4, 19, 42, 8, 31]

[38,16,27,4] [19,42,8,31]

Apply Merge Sort to the given sequence.

Complete the divide-and-merge tree until individual elements are reached, show the comparisons performed during each merge operation, and obtain the sorted sequence. Analyze the best-case and worst-case time complexity of Merge Sort.

12 a. A rod of length 8 units has the following selling prices: [CO4] [K3]

Length (i)	1	2	3	4	5	6	7	8
Price (p_i)	1	5	8	9	10	17	17	20

Formulate the recurrence relation for the Rod Cutting Problem and apply dynamic programming to determine the maximum obtainable revenue for a rod of length 8. Construct the DP table and identify the cuts that produce the optimal revenue.

[OR]

b. Consider the strings: [CO4] [K3]

X=ABCBDAB

Y=BDCABA

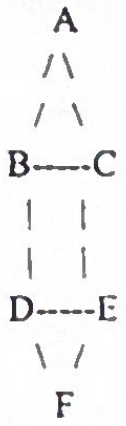
Apply the dynamic-programming algorithm for the Longest Common Subsequence problem. Determine the length of the LCS, and trace the created table to obtain one longest common subsequence.

13 a. Transform the string INTENTION into EXECUTION using insertion, deletion, and replacement operations, each having unit cost. Construct the recurrence relation and DP table for the String Editing / Edit Distance problem. Determine the minimum edit distance and analyze the sequence of editing operations represented by an optimal path through the table. [CO4] [K3]

[OR]

b. Consider the sequence: [CO4] [K3]
[5,3,4,8,6,7,9,7]
Analyze the sequence using dynamic programming to determine a Longest Non-Decreasing Subsequence (LNDS). Construct the required DP values for all positions, trace the elements forming an optimal subsequence, and determine the time complexity of the approach.

14 a. Consider the graph: [CO5] [K3]



Apply the backtracking technique to solve the graph-coloring problem using at most three colors.

[OR]

b. A knapsack has capacity (W=10). The available items are: [CO5] [K3]

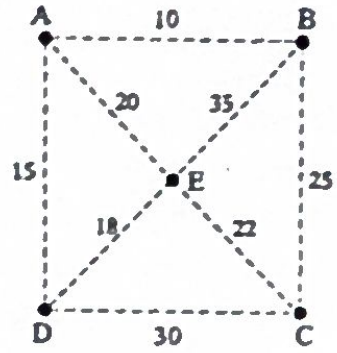
Item	Profit	Weight
1	40	2
2	30	5
3	50	10
4	10	5

Apply the Branch-and-Bound technique to solve the 0/1 Knapsack problem. Calculate suitable bounds at the generated nodes, identify the nodes that can be pruned, and determine the item combination giving the maximum profit.

15 a. Analyze the relationship among P, NP, NP-Complete, and NP-Hard problems. Discuss polynomial-time verification and polynomial-time reduction, and use suitable examples to justify how a problem may be classified as NP-Complete or NP-Hard. [CO5] [K4]

[OR]

b. A Traveling Salesperson instance consists of the following weighted graph: [CO5] [K3]



Analyze how an approximation algorithm can be used to obtain a near-optimal solution for the Traveling Salesperson Problem when finding an exact optimal tour is computationally expensive.